
Where Does Platonic Convergence Live? A Linear Readout Gains Where the Galaxy Embedding Bends Toward the Label and Loses Where It Bends Away

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Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations of galaxy images
3 with mutual k -nearest-neighbour agreement (MKNN). We ask whether that conver-
4 gence is concentrated where the embedding manifold is flat. A pointwise curvature
5 field from a trained decoder, validated against analytic truth, finds mean-curvature
6 magnitude correlating negatively with cross-survey neighbour agreement at every
7 latent dimension, and that local density removes most of it ($d = 20$), half ($d = 25$)
8 or little of an already small value ($d = 32$). A second-order expansion of a linear
9 map restricted to the manifold explains why: a readout sees the second fundamen-
10 tal form contracted with its normal component, $\langle w_N, \Pi \rangle$, an alignment’s second-
11 order error is a connection mismatch plus the relative second fundamental form,
12 and mean curvature retains only the metric trace of Π and determines neither. On
13 86,471 labelled galaxies a linear readout gains accuracy where the manifold bends
14 toward the label’s own curvature and loses it where the manifold bends away. Inter-
15 vening on the readout at a fixed manifold, the decoder’s second-order term raises
16 local R^2 at 79–99% of anchors, its mirror image lowers it at 97–100%, and a ran-
17 dom normal direction of the same norm does neither, on five encoders; across
18 anchors, the mismatch between the label Hessian and the probe-facing Hessian
19 is negative wherever the label has local structure and survives cross-fitting and a
20 change of encoder. Mean curvature of a single representation is therefore not a
21 viable explanation for weak Platonic convergence; the obstruction is relative and,
22 for the fitted global linear alignment, already present at first order.

23 1 Introduction

24 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data
25 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on the sky:
26 galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC) share only 0.3 to
27 2.3 percent of nearest neighbours across surveys under a family of vision models. We ask where on
28 the manifold that agreement lives: if two models disagree most where the embedding surface bends
29 sharply, local mean curvature should associate negatively with local cross-survey agreement.

30 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
31 trained decoder, validated against analytic truth in the production regime; (ii) a density-controlled
32 test of the curvature–alignment prediction, which finds no effect free of density; (iii) the residual
33 expansion of a linear map on a submanifold, which names the curvature such a map responds to
34 and shows mean-curvature magnitude is blind to it; and (iv) known-surface, cross-encoder and inter-
35 ventional tests on the readout side: a linear readout gains local accuracy where the manifold bends

36 toward the label’s curvature and loses it where the manifold bends away, at nearly every anchor on
 37 five encoders.

38 2 Pointwise curvature from a decoder

39 **Data.** DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both
 40 HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019], (Platonic Universe release [Du-
 41 raphe et al., 2025]); we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-
 42 survey ViT-B embeddings with catalogue labels (r -band magnitude, photometric redshift, smooth
 43 fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the readout experiment.
 44 Embeddings are unit-normalized (data on S^{767}). Curvature is evaluated only where J has full rank.

45 **Instrument.** Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden lay-
 46 ers of width 250, SiLU), fit on 80% of rows; curvature is evaluated on held-out rows. With
 47 $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental
 48 form $\Pi = P_N D^2 F(z)$, the mean-curvature vector of the raw decoder surface is $H_{\text{raw}} = \text{tr}_g \Pi$, by au-
 49 tomatic differentiation. On the unit sphere every d -submanifold has radial mean curvature $-d$, which
 50 says nothing about shape; differentiating $F/\|F\|$ instead splits H exactly as $-d\hat{x} + H^S$ ($H_{\text{rad}} = -d$
 51 to 10^{-14}), and H^S carries the shape. Section 3 uses the pre-registered $\|H_{\text{raw}}\|$; Sections 4–5 use the
 52 projected decoder.

53 **Validation.** Scored against closed-form curvature on rank (Spearman), direction (cosine) and mag-
 54 nitude ratio. At $D = 768$ an in-sphere surface $G(z) = Q \text{ normalize}([\phi(z); 0.8h(z); 0 \dots])$ (ϕ in-
 55 verse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h a scalar sum of four Gaussian bumps, Q a rotation; 18-dimensional
 56 span) at $n = 86,471$, $k = 2,048$ gives rank 0.94 against the true $\|H^S\|$, direction cosine 0.999 and
 57 magnitude ratio 1.00 for the full vector.

58 3 Curvature and cross-survey convergence

59 **Probe.** Per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k =$
 60 20 between the HSC and Legacy Survey embeddings, the Platonic Universe statistic made local,
 61 correlated with $\|H_{\text{raw}}\|$ over all 10,000 points at $d \in \{20, 25, 32\}$.

62 **Result.** The raw association $\rho(\|H_{\text{raw}}\|, \text{MKNN})$ is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$
 63 ($p < 10^{-5}$, $< 10^{-5}$, 0.009). But k -NN statistics are density-sensitive [Radovanović et al., 2010]:
 64 local density ($k = 30$ estimate on the Legacy embedding) correlates with agreement at -0.21 at
 65 every d and with $\|H_{\text{raw}}\|$ at $+0.43$, $+0.32$ and $+0.01$, so residualizing on density leaves -0.024 ,
 66 -0.066 and -0.022 (78%, 49% and 8% removed). Under a permutation null within density strata
 67 only $d = 25$ survives ($p < 5 \times 10^{-5}$; three decoder seeds; CKA [Kornblith et al., 2019] agrees), and
 68 removing the radial component of H_{raw} collapses it to -0.023 , into the range of the two d that fail.

69 4 What a linear map sees on a curved manifold

70 Let $M \subset \mathbb{R}^D$ be a d -dimensional submanifold, $d < D$; at x_0 take an orthonormal tangent frame J ,
 71 the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr} \Pi$, and for a vector w its
 72 tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$,
 73 so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

74 To second order, a linear function is as non-linear on M as M bends in that function’s normal direc-
 75 tion [Lee, 2018]. For $M \subset S^{D-1}$ the normal bundle splits into the sphere’s radial direction and the
 76 rest, $\Pi = \Pi^S - g\hat{x}$, so $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x})g$ (2): a linear readout of a unit-normalized
 77 embedding is bent by the manifold’s shape and, whatever the manifold, by the sphere in proportion
 78 to the readout’s own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and
 79 a patch $u \sim N(0, s^2 I_d)$ in normal coordinates, embedding and label locally C^4 (the $\frac{1}{2}$ is the Gaus-
 80 sian constant; real k -NN patches share only the tensor structure), with $\bar{e}$ the patch-mean residual,

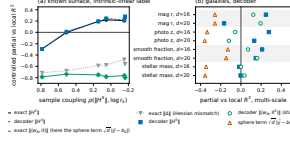


Figure 1: (a) Known surface, intrinsically linear label, exact geometry: three-control partial against local R^2 as the density-curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$, multi-scale control: the decoder’s $\|H^S\|$ (filled squares), shape term (open circles), sphere term (open triangles).

label	$\ H^S\ $		shape $\ \langle w_N, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control; alignment is $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$. * not significant at 0.05; the three-control partials agree in sign in every significant cell.

$$\delta = [\nabla_M y]_J - J^\top w, \Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle, \quad \mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_F^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

For two embeddings F, G of the same objects in a common chart u and a linear map A with $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2} [J_G(\Gamma^G - \Gamma^F) + \Pi_G - A\Pi_F](u, u) + O(|u|^3)$, with Γ the induced connections: the second-order obstruction to linear alignment is a connection mismatch plus the *relative* second fundamental form, and reduces to the latter when the induced connections agree at x_0 ; the extrinsic term’s normal part is $\Pi_G - P_N^G A \Pi_F$.

So $\|H\|$, the norm of the metric trace of Π , need not correlate with either $\langle w_N, \Pi \rangle$ or the relative term; the second-order term scales as s^4 and, with fixed- k neighbourhoods, density sets s , so curvature effects scored on patches are density-sensitive.

5 Probe-facing curvature: bending toward the label helps, bending away hurts

Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered $\alpha = 100$ from embedding to label; local R^2 inside each of 512 anchors’ 2,048-neighbourhoods; rank-partial Spearman against local R^2 with Freedman-Lane p -values [Freedman and Lane, 1983], control-ling for log patch radius, local label variance and count and, stronger, log radius at five scales $k \in \{16, \dots, 2048\}$. Per anchor we compute $\|H^S\|$ and the probe-facing Hessian $\|\langle w_N, \Pi \rangle\|_g$ with $w_N = w - Jg^{-1}J^\top w$, split by Eq. (2) into its shape and sphere terms.

Known surface. The validation surface resampled with weight $\|H^S\|^\gamma$, so only where the points sit changes (Figure 1a): the $\|H^S\|$ partial follows the sampling from -0.29 to $+0.22$, reproduced by the decoder, while the probe-facing Hessian $\|\langle w_N, \Pi \rangle\|$ holds at -0.74 to -0.80 at every sampling, there entirely the sphere term of Eq. (2) (rank 0.996–1.000 with $\sqrt{d} |\hat{y} - b_0|$).

Galaxies. Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank 0.96–0.99 with the full norm). We estimate $\text{Hess}_M y$ per anchor by a least-squares quadratic of the label on the tangent-projected coordinates $u = g^{-1}J^\top(x - x_0)$ of its 2,048 neighbours (a local Monge chart, so the quadratic coefficient is the covariant Hessian). Three regularities (Table 1; other decoder seeds and widths in Appendix A). The sphere term is negative on every label and both d (-0.13 to -0.36) under the sealed probe; with a weak-ridge probe ($\alpha = 1$, Appendix B) it stays negative for magnitude and smooth fraction (-0.21 , -0.12) and vanishes for redshift and stellar mass, where the sealed-probe sign came with ridge shrinkage of extreme predictions. Wherever

the label has local structure (magnitude, redshift, smooth fraction; stellar mass is indistinguishable from zero throughout) the mismatch is negative (-0.09 to -0.45 ; unchanged under cross-fitting, Appendix B) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$). The shape term alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift); with $K = \langle w_N, \Pi \rangle$ it is one of three terms in $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, K \rangle + \|K\|^2$, so raising it shrinks or enlarges the mismatch with the sign of the cross term. The same regression on the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median cosine with the decoder’s 0.55 – 0.65).

Bending toward helps, bending away hurts. By Eq. (3) curvature lowers the second-order residual relative to a flat patch iff $2\langle \text{Hess}_M y, K \rangle > \|K\|^2$: the manifold must bend toward the label’s own curvature, and not too far. (For least squares on a hypersurface the optimal normal weight is the label’s second derivatives contracted with the principal curvatures over their squares [Liu et al., 2023], derived there as a uniqueness and stability condition; here it is measured per anchor on learned embeddings.) The partials compare anchors, and no anchor is flat, so we test the condition by intervening on the readout at a fixed manifold. At each anchor the fitted weight is split into tangent, sphere and in-sphere normal parts, and the in-sphere normal part is replaced by t times the decoder’s second-order term $\frac{1}{2}\langle w_S, \Pi^S \rangle(u, u)$: $t = 1$ is the manifold’s own bending, $t = -1$ its mirror image, $t = 0$ flat. On five encoders and all four labels, $t = 1$ beats flat at 79–99% of anchors (median local R^2 gain $+0.013$ to $+0.031$), $t = -1$ is worse than flat at 97–100% (-0.025 to -0.055), and a random normal direction of the same norm does neither (Appendix E). The optimum sits at t^* of 1.5 to 3.6 : the ridge-shrunk readout under-uses the bending, consistent with the weak-ridge gain above.

Across encoders. The quadratic-chart estimator’s mean-bending trace at the same anchors of five encoders (ViT-B, DINOv3, CLIP-B, ConvNeXt-B, ViT-L; $d = 16$) has density-controlled associations with the magnitude probe’s local R^2 of both signs (-0.24 to $+0.27$), as Eq. (1) allows. The decoder test repeated per encoder (Appendix D) is encoder-stable where the expansion says it should be: the mismatch partial is negative for magnitude and redshift on all five encoders (-0.32 to -0.61 ; cross-fitted -0.28 to -0.62) and the alignment positive for magnitude on all five ($+0.13$ to $+0.53$) and for redshift on four; the shape term’s sign varies with encoder as well as label; stellar mass stays null but for one mismatch cell (-0.12).

6 Discussion

A linear readout of a galaxy embedding gains accuracy where the manifold bends toward the label’s curvature and loses it where the manifold bends away, at nearly every anchor and on five encoders. The residual expansion names the quantity, $\langle w_N, \Pi \rangle$ against $\text{Hess}_M y$, and a decoder’s pointwise second fundamental form measures it at production scale, agreeing with analytic truth and with the data. The expansion’s other terms behave as predicted (mismatch negative and robust to cross-fitting, probe strength and encoder; sphere term negative on two of four labels under a weak ridge; shape term label-signed through its alignment with the label’s Hessian). The same expansion explains why mean-curvature statistics disagree across estimators and encoders: they are traces of a tensor that each map contracts differently. For a linear alignment map the relevant second-order quantity is not one embedding’s mean curvature but $\Pi_G - P_N^G A \Pi_F$, which needs both second fundamental forms and a map between tangent frames. Our pilot (Appendix C) finds a 44% first-order residual in carrying HSC tangent frames into the Legacy embedding under the fitted global ridge map, and no estimator-stable second-order association. On this survey pair, single-model mean curvature has no robust density-controlled association with cross-survey convergence.

Limitations. One survey pair. The label’s Hessian is a local estimate (R^2 gain over linear 0.10 – 0.17); its norm is reliably ranked (split-half ρ 0.92) but its direction only moderately (tensor cosine 0.19 – 0.35). On a synthetic surface with known curvature the mismatch partial reproduces (-0.14 to -0.88) but the alignment partial has no stable sign; Appendix E, not the fixture, ties alignment to local R^2 . Stars are per cell; mismatch and alignment are the primary hypotheses, the rest exploratory. Permutation p -values assume anchor exchangeability under overlapping patches.

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190 Appendix A: decoder ablations

191 Table 2 repeats the four quantities of Table 1 at $d = 16$ under the multi-scale density control for two
192 further decoder initialisation seeds and for a decoder of width 400^3 instead of 250^3 (all fits reach
193 variance explained 0.952). Every significant sign of the main table is retained.

194 Appendix B: sensitivity of the mismatch and sphere columns

195 Cross-fitting: each 2,048-patch is split at random into halves; $\text{Hess}_M y$ is fitted on one half and local
196 R^2 scored on the other (both directions shown), so the Hessian estimate and the outcome share no
197 rows. The split-half cosine between the two Hessian estimates is the reliability of the estimate. Weak
198 ridge: the probe refit at $\alpha = 1$ instead of 100 (global out-of-sample R^2 rises by 0.10 to 0.15), which
199 substantially reduces the shrinkage of extreme predictions; the sphere term is recomputed from that
200 probe.

201 Appendix C: relative second fundamental form, pilot

202 Two sphere-projected decoders (HSC = F , Legacy = G ; Phase 7 protocol), a global ridge map A
203 from x_F to x_G (fit on the 8,000 training rows), 2,048 seeded anchors. Columns: holdout R^2 of A ;
204 median first-order obstruction $\|AJ_F - J_G L\|/\|AJ_F\|$ with $L = J_G^+ AJ_F$; then multi-scale density-
205 controlled partials against MKNN ($k = 20$; log radius at $k \in \{10, 30, 100, 300\}$ in both spaces) of
206 each decoder’s $\|H^S\|$, the first-order obstruction, $\|\text{II}_G(L \cdot, L \cdot) - P_N^G A \text{II}_F\|$, the same with L fitted
207 on 256 neighbours’ latent codes, and a decoder-free estimate from the quadratic coefficient of the
208 alignment residual on F ’s tangent coordinates.

label	quantity	seed 0	seed 1	seed 2	400 ³
mag_r	shape	+0.12	+0.05*	+0.12	+0.05*
	sphere	−0.26	−0.26	−0.27	−0.26
	mismatch	−0.39	−0.35	−0.34	−0.34
	alignment	+0.35	+0.31	+0.37	+0.41
photo_z	shape	−0.06*	−0.09*	−0.09	−0.06*
	sphere	−0.33	−0.35	−0.34	−0.35
	mismatch	−0.45	−0.43	−0.42	−0.41
	alignment	+0.26	+0.24	+0.25	+0.25
smooth_fraction	shape	+0.07*	+0.20	+0.15	+0.10
	sphere	−0.13	−0.15	−0.14	−0.11
	mismatch	−0.09*	−0.05*	−0.08*	−0.07*
	alignment	+0.19	+0.14	+0.09	+0.10
stellar_mass	shape	+0.00*	−0.06*	+0.05*	+0.09
	sphere	−0.35	−0.35	−0.35	−0.35
	mismatch	−0.04*	+0.03*	+0.02*	+0.03*
	alignment	+0.01*	−0.02*	+0.03*	−0.02*

Table 2: Decoder ablations, $d = 16$, 512 anchors, multi-scale density control. * not significant at 0.05.

label	d	mismatch, cross-fit		alignment, cross-fit		Hess. split cos	sphere, $\alpha = 1$
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	p50	(main: $\alpha = 100$)
mag_r	16	−0.38	−0.34	+0.33	+0.31	+0.35	−0.21 (−0.26)
mag_r	20	−0.43	−0.40	+0.39	+0.37	+0.24	−0.22 (−0.26)
photo_z	16	−0.42	−0.36	+0.23	+0.20	+0.32	+0.05* (−0.33)
photo_z	20	−0.42	−0.42	+0.22	+0.22	+0.20	+0.03* (−0.36)
smooth_fraction	16	−0.10	−0.17	+0.17	+0.19	+0.33	−0.12 (−0.13)
smooth_fraction	20	−0.15	−0.21	+0.05*	+0.03*	+0.23	−0.13 (−0.15)
stellar_mass	16	+0.01*	−0.04*	−0.03*	+0.06*	+0.31	−0.02* (−0.35)
stellar_mass	20	−0.06*	−0.07*	+0.01*	−0.00*	+0.19	−0.02* (−0.35)

Table 3: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

Appendix D: the same test on four further encoders

Table 5 repeats Table 1 at $d = 16$ on the same 86,471 galaxies embedded by four further encoders from the same release (DINOv3 ViT-B/16, CLIP ViT-B, ConvNeXt-B, ViT-L; widths 768, 512, 1,024, 1,024; each unit-normalized), with a decoder of the same width and protocol fitted per encoder (seed 0; variance explained 0.966, 0.985, 0.965, 0.968 for the four new fits), 512 anchors, $k = 2,048$, multi-scale density control. The first row per label is the probe’s global out-of-sample R^2 ; Table 6 gives the cross-fitted mismatch and alignment and the split-half reliability of the label Hessian per encoder.

Appendix E: intervening on the readout’s normal component

The partials of Table 1 compare anchors with one another and cannot say whether bending away from the label’s curvature hurts relative to no bending, since no anchor is flat. Here the readout is changed at a fixed manifold. At each anchor the fitted ridge weight is split into tangent, radial and in-sphere normal parts $w = w_T + w_{\text{rad}} + w_S$, and the readout $w_T + w_{\text{rad}} + t q$ is scored on the anchor’s 2,048 neighbours with the local intercept refit, where q is the decoder’s second-order term $\frac{1}{2}\langle w_S, \Pi^S \rangle(u, u)$ in the anchor’s chart coordinates: $t = 1$ is the manifold bending as it does in the probe’s normal direction, $t = -1$ the mirror-image bending, $t = 0$ flat. The local sum of squares is an exact quadratic in t with minimum t^* . Columns: fraction of anchors where $t = 1$ beats flat (help) and where $t = -1$ is worse than flat (hurt), the median change in local R^2 at $t = \pm 1$, the median t^* ; then the same for a random in-sphere normal direction of the same norm as w_S .

run	R_A^2	tan. resid	$\ H^S\ $ HSC	$\ H^S\ $ Legacy	tan. resid	Π_{rel}	Π_{rel} (local L)	Π_{rel} (data)
$d = 20$, seed 0	0.64	0.44	−0.15	+0.04*	−0.18	+0.04*	−0.18	+0.13
$d = 25$, seed 0	0.64	0.43	−0.10	+0.06	−0.16	+0.05	−0.23	+0.13
$d = 20$, seed 1	0.64	0.44	−0.11	+0.05	−0.19	+0.04	−0.20	+0.14

Table 4: Relative-II pilot. * not significant at 0.05.

label	quantity	ViT-B	DINOv3	CLIP-B	ConvNeXt-B	ViT-L
mag_r	global R^2	0.52	0.57	0.50	0.52	0.52
	shape	+0.12	−0.11	+0.16	+0.16	−0.21
	sphere	−0.26	−0.23	−0.40	−0.24	−0.08*
	mismatch	−0.39	−0.57	−0.34	−0.32	−0.44
	alignment	+0.35	+0.53	+0.32	+0.21	+0.13
photo_z	global R^2	0.51	0.56	0.49	0.49	0.51
	shape	−0.06*	−0.05*	−0.17	+0.29	−0.16
	sphere	−0.33	−0.21	−0.29	−0.46	−0.44
	mismatch	−0.45	−0.61	−0.45	−0.48	−0.46
	alignment	+0.26	+0.33	+0.06*	+0.22	+0.14
smooth_fraction	global R^2	0.53	0.59	0.49	0.56	0.55
	shape	+0.07*	−0.08*	+0.20	+0.24	+0.06*
	sphere	−0.13	−0.06*	+0.05*	−0.15	−0.11
	mismatch	−0.09*	−0.33	−0.25	−0.20	−0.01*
	alignment	+0.19	−0.02*	+0.24	+0.04*	+0.14
stellar_mass	global R^2	0.48	0.54	0.50	0.49	0.49
	shape	+0.00*	−0.15	−0.10	−0.11	−0.13
	sphere	−0.35	−0.26	−0.38	−0.40	−0.38
	mismatch	−0.04*	+0.03*	−0.12	−0.06*	−0.02*
	alignment	+0.01*	−0.05*	−0.01*	+0.07*	−0.03*

Table 5: Cross-encoder replication, $d = 16$, multi-scale density control. * not significant at 0.05.

encoder	label	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	
DINOv3	mag_r	−0.60	−0.62	+0.50	+0.50	+0.35
	photo_z	−0.56	−0.59	+0.28	+0.27	+0.24
	smooth_fraction	−0.32	−0.29	−0.03*	−0.02*	+0.31
	stellar_mass	+0.03*	+0.02*	−0.03*	−0.07*	+0.22
CLIP-B	mag_r	−0.28	−0.35	+0.28	+0.29	+0.31
	photo_z	−0.41	−0.40	+0.08*	+0.06*	+0.25
	smooth_fraction	−0.22	−0.24	+0.15	+0.20	+0.28
	stellar_mass	−0.14	−0.13	−0.04*	+0.01*	+0.19
ConvNeXt-B	mag_r	−0.36	−0.34	+0.19	+0.21	+0.29
	photo_z	−0.46	−0.45	+0.18	+0.19	+0.30
	smooth_fraction	−0.15	−0.19	−0.02*	+0.03*	+0.32
	stellar_mass	−0.05*	−0.01*	+0.14	−0.03*	+0.29
ViT-L	mag_r	−0.46	−0.49	+0.09*	+0.15	+0.30
	photo_z	−0.44	−0.42	+0.09*	+0.13	+0.24
	smooth_fraction	−0.03*	−0.04*	+0.11	+0.07*	+0.30
	stellar_mass	−0.06*	+0.01*	−0.04*	−0.03*	+0.21

Table 6: Cross-fitted mismatch and alignment partials per encoder (multi-scale control) and split-half reliability of the label Hessian.

run	label	decoder second-order term					random normal direction				
		help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	t^*	help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	
ViT-B, $d = 16$	mag_r	0.97	1.00	+0.030	-0.051	2.3	0.28	0.71	-0.006	-0.006	
	photo_z	0.99	0.99	+0.028	-0.038	3.0	0.38	0.70	-0.003	-0.004	
	smooth_fraction	0.90	1.00	+0.021	-0.040	1.7	0.24	0.79	-0.008	-0.009	
	stellar_mass	0.96	0.99	+0.026	-0.040	2.6	0.34	0.68	-0.004	-0.005	
ViT-B, $d = 20$	mag_r	0.99	1.00	+0.031	-0.049	2.6	0.30	0.72	-0.005	-0.006	
	photo_z	0.99	1.00	+0.029	-0.038	3.6	0.37	0.68	-0.002	-0.004	
	smooth_fraction	0.90	1.00	+0.020	-0.035	1.9	0.26	0.77	-0.007	-0.008	
	stellar_mass	0.97	1.00	+0.027	-0.038	3.1	0.31	0.72	-0.003	-0.004	
DINOv3, $d = 16$	mag_r	0.92	0.99	+0.018	-0.031	2.0	0.36	0.70	-0.003	-0.004	
	photo_z	0.96	0.99	+0.020	-0.030	2.6	0.36	0.62	-0.002	-0.002	
	smooth_fraction	0.93	1.00	+0.024	-0.055	1.5	0.28	0.72	-0.009	-0.008	
	stellar_mass	0.95	1.00	+0.016	-0.028	2.1	0.32	0.64	-0.002	-0.002	
CLIP-B, $d = 16$	mag_r	0.95	0.99	+0.018	-0.031	2.2	0.34	0.66	-0.004	-0.004	
	photo_z	0.99	1.00	+0.019	-0.026	3.4	0.39	0.62	-0.002	-0.002	
	smooth_fraction	0.92	0.99	+0.014	-0.027	1.7	0.28	0.69	-0.004	-0.004	
	stellar_mass	0.97	1.00	+0.020	-0.028	2.8	0.36	0.65	-0.002	-0.002	
ConvNeXt-B, $d = 16$	mag_r	0.92	1.00	+0.018	-0.031	2.0	0.32	0.64	-0.004	-0.003	
	photo_z	0.99	1.00	+0.018	-0.026	2.9	0.36	0.59	-0.002	-0.001	
	smooth_fraction	0.88	0.98	+0.016	-0.036	1.6	0.31	0.69	-0.004	-0.004	
	stellar_mass	0.99	1.00	+0.022	-0.032	3.1	0.38	0.57	-0.002	-0.001	
ViT-L, $d = 16$	mag_r	0.96	0.99	+0.022	-0.034	2.6	0.32	0.69	-0.003	-0.004	
	photo_z	0.97	0.99	+0.018	-0.025	3.2	0.40	0.61	-0.002	-0.002	
	smooth_fraction	0.79	0.97	+0.013	-0.029	1.6	0.24	0.77	-0.005	-0.007	
	stellar_mass	0.98	1.00	+0.022	-0.032	2.9	0.35	0.64	-0.002	-0.002	

Table 7: Counterfactual scaling of the readout’s in-sphere normal component, 512 anchors per run. Bending as the manifold does helps at nearly every anchor, the mirror-image bending hurts at nearly every anchor, a random normal direction does neither; $t^* > 1$ throughout, so the ridge-shrunk normal component undershoots.